ROUGH DRAFT

Beneath the Mesa

Assessing Aquifer and Well Water Resources Across Sandoval County, New Mexico

A County-Scale Screening Study with Cuba, New Mexico as a Focus Area

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ABSTRACT
Water scarcity is a persistent challenge across New Mexico. Sandoval County spans several hydrogeologic settings, and its communities depend on a mix of groundwater wells, surface water, and acequia irrigation systems. This project focuses on groundwater: it evaluates the condition of the county's aquifers and wells, with Cuba, New Mexico, on the eastern edge of the San Juan Basin, as a focused case study. The project is motivated by concern about declining groundwater near Cuba, but rather than assume how severe that decline is or what is causing it, we test what the available public data actually show, and where they are too sparse to say. We combined public water-level, well, aquifer-test, geologic, and streamflow records from the USGS, the New Mexico Bureau of Geology, and the State Engineer, and analyzed them with reproducible code. Five of the nine long-term monitoring wells, all 65–95 km from Cuba, fell over their 1998–2026 records, while the Sandia site north of Albuquerque has risen about 13 ft since 2007. Near Cuba, the only well records are one-time aquifer tests from 1957–1979, and no well within 25 km has been measured repeatedly, so the data can neither confirm nor rule out severe depletion of Cuba's aquifer. Answering that question requires the village's own well records.

1 Introduction

1.1 Background and Motivation

[Background on water resources in New Mexico and Sandoval County, and the concern about declining groundwater near Cuba that motivated the project.]

1.2 Approach

The study works at two levels:

- **County-wide screening.** Map and compare aquifers, wells, and groundwater levels across Sandoval County. Because the county crosses several geologic regions, comparisons are grouped by area and, where the records name it, by aquifer unit, instead of treating the county as one uniform aquifer.
- **Cuba case study.** A closer look at the Ojo Alamo Sandstone and San Jose Formation near Cuba, using published maps of both formations, aquifer-test records from nearby wells, and stream measurements on the Rio Puerco at Cuba.

1.3 Research Questions

RQ1. How have groundwater levels changed at monitored wells in Sandoval County, and which wells show sustained decline, stability, or recovery?

RQ2. What do well records show about well depth, depth to water, and the aquifer units that wells in different parts of the county draw from?

RQ3. What do available records show about groundwater near Cuba, and are they enough to evaluate the concern that its main aquifer is approaching severe depletion?

RQ4. How do the Ojo Alamo Sandstone and San Jose Formation compare near Cuba in extent, thickness, and hydraulic properties?

RQ5. Where is monitoring coverage strong, and where are the most important data gaps?

RQ6. Which indicators can be combined into a dashboard that shows groundwater stress without overstating how certain the data are?

The project proposal also asked about groundwater qual-

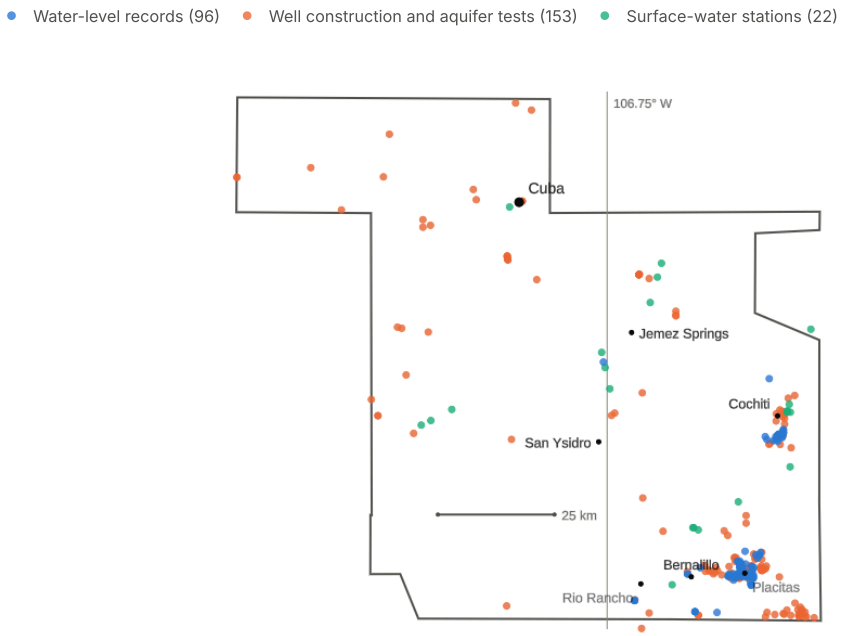


Figure 1. Every mapped site in the project's datasets, colored by what it measures: water-level records (USGS monitoring wells and NMBGMR wells), well construction and aquifer tests (OSE aquifer tests and well logs), and surface-water stations (USGS stream, canal, and reservoir sites, including the Rio Puerco at Cuba). Most sites lie east of 106.75° W along the Rio Grande; within 25 km of Cuba there are only one-time aquifer tests, one well log, and one stream gage. A dot is a site with any record, not necessarily active monitoring. Outline: Sandoval County (simplified, US Census Bureau). Town locations approximate.

ity, acequias, and links between groundwater and surface water. Those questions were dropped because the data needed to answer them were not available (Section 7).

2 Study Area

2.1 Sandoval County

[Describe the county, its hydrogeologic settings, and its main communities and water sources (Fig. 1).]

2.2 Cuba Focus Area

Cuba lies on the eastern edge of the San Juan Basin. The case study focuses on the Ojo Alamo Sandstone and San Jose Formation near Cuba (U.S. Geological Survey, 2020; Kernodle, 1996). In this report, the *Cuba area* is the area within 25 km of the village center; distances to Cuba are straight-line distances from that center. *[Describe Cuba, its water supply, and the aquifers it draws from.]*

3 Data

Town-specific data for Cuba are limited, so the study relies on public datasets that can be filtered to Sandoval County and the area around Cuba. Eight datasets were collected (Table 1). For seven of them, a one-page brief records what the dataset contains, which research questions it can help answer, which plots to make from it, and what it can't show; a plotting guide compares all seven. The raw files are kept unedited in the project repository (Djordjevic et al., 2026).

3.1 Datasets Used

Groundwater levels. Two datasets measure depth to water below land surface, where a larger number means a lower water table. The USGS statistics (U.S. Geological Survey, 2026c) give long monthly records for 9 wells at four sites: Sandia, Lincoln Middle School in Rio Rancho, Bernalillo, and Dome Road. The same download also covers stream gages, canals, and reservoirs, but only the groundwater records are used. The NM Bureau of Geology measurements (New Mexico Bureau of Geology and Mineral Resources, 2026c) cover more wells but fewer years: most were measured in 1997–1998, and only 5 wells were measured again in 2016–2019, so few of them can show a long-term trend.

Well and aquifer records. The State Engineer's aquifer-test records (New Mexico Office of the State Engineer, 2026b) are the only dataset with hydraulic properties, such as transmissivity and storativity, and they name the formation for 125 of the 179 tests. Seven test events name the Ojo Alamo Sandstone, among them 1957 and 1960 tests at what may be a single school well 0.7 km from Cuba, but only one names the San Jose Formation.

Table 1. Datasets collected for the study. Distances are measured from Cuba.

Dataset	Source	Contents	Relevance to Cuba	Used for
Groundwater-level statistics	USGS	Monthly and yearly depth to water at 9 wells on 4 sites, 1998–2026	Wells are 65–95 km away, in the eastern part of the county	RQ1, RQ5
Groundwater levels	NM Bureau of Geology	456 depth-to-water measurements at 87 wells, 1997–2019, with well depth	Nearest well is 38 km away	RQ1, RQ2, RQ5
Aquifer tests	NM Office of the State Engineer	179 tests in Sandoval County, 1930–2008: well construction, formation, and hydraulic properties	Includes tests at a well 0.7 km from Cuba	RQ2–RQ4
Wells with well logs	NM Office of the State Engineer	Drilling records for 6 Sandoval County wells, 1978–2022: borehole depth and depth to first water	Link to Cuba's wells not verified	RQ2, RQ5
Ojo Alamo Sandstone	USGS map atlas, 1990	Maps of depth to the top of the sandstone, its thickness, and wells completed in it	Regional; covers the Cuba area	RQ4
San Jose Formation near Lindrith	NM Geological Society, 2025	Water-bearing units, thickness of the Cuba Mesa Member, and groundwater levels	Centered near Lindrith, about 30 km north	RQ4
Rio Puerco at Cuba	USGS station 08332525	10 streamflow measurements (2019–2026) and 4 gage-height readings	At Cuba, but measures the river, not groundwater	RQ3, RQ5
Aquifer sensitivity	NM Water Data	Sensitivity class for 613 mapped areas	Can't be located: the file has no coordinates	Not used

The State Engineer's well logs (New Mexico Office of the State Engineer, 2026c) add borehole depth and the depth at which water was first found on the day each well was drilled, which is not a current water level.

Geology of the Cuba-area aquifers. Two publications describe the formations studied for RQ4: a USGS map atlas of the Ojo Alamo Sandstone across the San Juan Basin (Thorn et al., 1990), and a study of the San Jose Formation near Lindrith that identifies four water-bearing units and maps the Cuba Mesa Member (Blandford et al., 2025). Both are maps and reports rather than data tables, and neither shows which formation Cuba's municipal wells draw from.

Surface water at Cuba. USGS station 08332525 on the Rio Puerco (U.S. Geological Survey, 2026b) is the only dataset measured at Cuba itself. Six of its 10 streamflow visits found no flow, and the other four measured 6.9 to 20.4 cubic feet per second. Because these are occasional visits rather than a continuous record, the data are used only as context for the Cuba case study.

Not used. The aquifer sensitivity file (New Mexico Water Data, 2026) has no coordinates or shapes, so it can't be mapped or matched to Cuba without its separate map layer, which was not obtained.

3.2 Data Not Available

The most important missing data are Cuba's municipal well records: well IDs, the formation each well draws from, pumping volumes, and water-level history, held by the Village of Cuba or the State Engineer (New Mexico Office of the State Engineer, 2026a). Every dataset brief identified these records as the missing piece for RQ3. Also not obtained were the USGS digital aquifer data for the San Juan Basin (U.S. Geological Survey, 2020), groundwater-quality records such as the Brackish Water Assessment (New Mexico Bureau of Geology and Mineral Resources, 2026b), acequia locations (New Mexico Interstate Stream Commission, 2026), and records from other monitoring networks (U.S. Geological Survey, 2026a; New Mexico Bureau of Geology and Mineral Resources, 2026a). Section 7 describes what their absence means for the findings.

Table 2. Cleaned files produced so far.

File	Rows	Contents
groundwater_levels_monthly.csv	2,673	Monthly mean depth to water at 9 wells, June 1998 to September 2026
aquifer_tests.csv	179	One row per aquifer test in Sandoval County, 1930 to 2008, with 17 columns

4 Methods

4.1 Data Cleaning

Cleaning makes the data correct and consistent without drawing conclusions; averaging, trends, and narrowing to the Cuba area happen in the analysis. Raw downloads are kept unedited. A MATLAB script, `cleaning/clean_csvs.m`, reads them and writes the cleaned files (Table 2), so anyone can rerun it and get the same result. As it runs, it prints checks such as row counts, duplicate rows, and the list of formation names.

Shared conventions. Every cleaned file starts with the same six columns: site ID, site name, latitude, longitude, date, and source. Each has one row per mea-

surement, sorted by site and date. Numbers are stored without units or commas, dates are written as YYYY-MM-DD, and column names end with their unit, such as `well_depth_ft`. Missing values are left blank rather than filled with placeholders such as 0 or −9999, which a chart would plot as real values.

Groundwater levels. From the USGS statistics, the script keeps the depth-to-groundwater rows that are monthly means and renames the columns to the shared names. Wells with an A or B suffix are kept as separate sites, because they share a location but are measured at different depths. The cleaned file was checked against the raw data: it has the same rows, dates, and values, with no duplicate site-and-date rows.

Aquifer tests. The script keeps all 179 Sandoval County tests, including repeat tests on the same well, and the 16 raw columns the analysis needs: well construction, formation, water level, pumping rate, and hydraulic properties. Empty and administrative columns are dropped. It converts numbers stored as text, standardizes dates, and cleans formation names by trimming spaces, converting them to uppercase, and fixing a misspelling. Three decisions were needed:

- One hydraulic conductivity was recorded as a range, 11 to 21 ft/day; the midpoint, 16 ft/day, is used.
- 104 tests list January 1, 1900, as the date of the starting water level, meaning the date is unknown, so that column is left out.
- 30 tests are dated January 1, which may mean only the year is known; they are kept as is.

Many tests are missing values, such as the formation (54 tests) and storativity (165 tests); these cells are left blank.

Open issues. Most of the State Engineer's well IDs in the aquifer-test file are incomplete: 157 of the 179 are partial, so the cleaned file has 73 site IDs for about 149 distinct wells. The analysis therefore tells wells apart by name and location rather than by ID. The Bureau of Geology water levels, the Rio Puerco measurements, and the well logs still need cleaning; the first and last of these also need their UTM coordinates converted to latitude and longitude. *[Update this paragraph once the remaining files are cleaned.]*

4.2 Analysis

The analysis focuses on measurable indicators, such as depth to water, water-level change over time, well depth, aquifer unit, and hydraulic properties from aquifer tests, rather than a full hydrologic model. Where records are sparse, the results say so instead of filling the gaps. All analysis code is in the `analysis/` folder of the repository. Each chart is a Python function that returns the figure together with its takeaway, caption, scope, and labeled findings, so the app, this report, and the documentation show the same numbers; every chart is reproduced on the documentation site's Analysis page. Datasets not yet cleaned (NMBGMR water levels, the Rio Puerco visits, and the OSE well logs) receive only minimal preparation in code, such as converting UTM zone 13 coordinates to latitude and longitude; the raw files are never changed.

Water-level trends (RQ1). Monthly mean depths to water were averaged into annual means, using only years with at least 9 months of data. Each well's trend is Sen's slope (the median of all pairwise slopes), and its significance is the Mann–Kendall test; a well is called falling or rising only when $p < 0.05$ (Helsel et al., 2020). Trends are reported for the full record and for 2016–2025. Because annual levels are serially correlated, the p-values and intervals are optimistic.

Wells and aquifer units (RQ2, RQ4). Each well contributes one point: the median NMBGMR reading, the first OSE aquifer test, or the depth where water was first found in a well log. Wells were grouped by the nearest of seven towns. Aquifer-test properties were compared by recorded formation, using one median per well. The scanned USGS map of the Ojo Alamo Sandstone (Thorn et al., 1990) was read by eye, not digitized.

Cuba case study (RQ3). The "Cuba area" is defined as within 25 km of the village center. For every dataset, we measured the distance from Cuba to its nearest site and recorded what kind of record it holds.

Coverage and indicators (RQ5, RQ6). Every site in every dataset was mapped and counted by year to show where and when monitoring exists, and each dataset was rated against each research question from its brief. Indicators were kept only where they can be computed from the data, each with the place it applies to and a stated confidence.

4.3 Classifying Findings

Each finding is labeled as one of the following:

- **Measured:** a trend or pattern the data show directly.
- **Hypothesis:** a possible explanation the data are consistent with but don't prove.
- **Insufficient data:** a question the available records can't answer yet.

4.4 Interactive Dashboard

Results are presented in an interactive Streamlit dashboard built with pandas and Plotly. Code and documentation are maintained on GitHub (Djordjevic et al., 2026). The dashboard uses the same chart functions as this report. Besides a home page, it has five pages: a county map of every site; the groundwater-level records and trends (RQ1); well depth, depth to water, and aquifer properties by area and formation (RQ2, RQ4); the Cuba case study (RQ3, RQ4); and data sources with monitoring coverage and the groundwater-stress indicators (RQ5, RQ6). Each chart is shown with its takeaway, its scope (Cuba, the Cuba area, or Sandoval County), a caption, its labeled findings, and the data behind it,

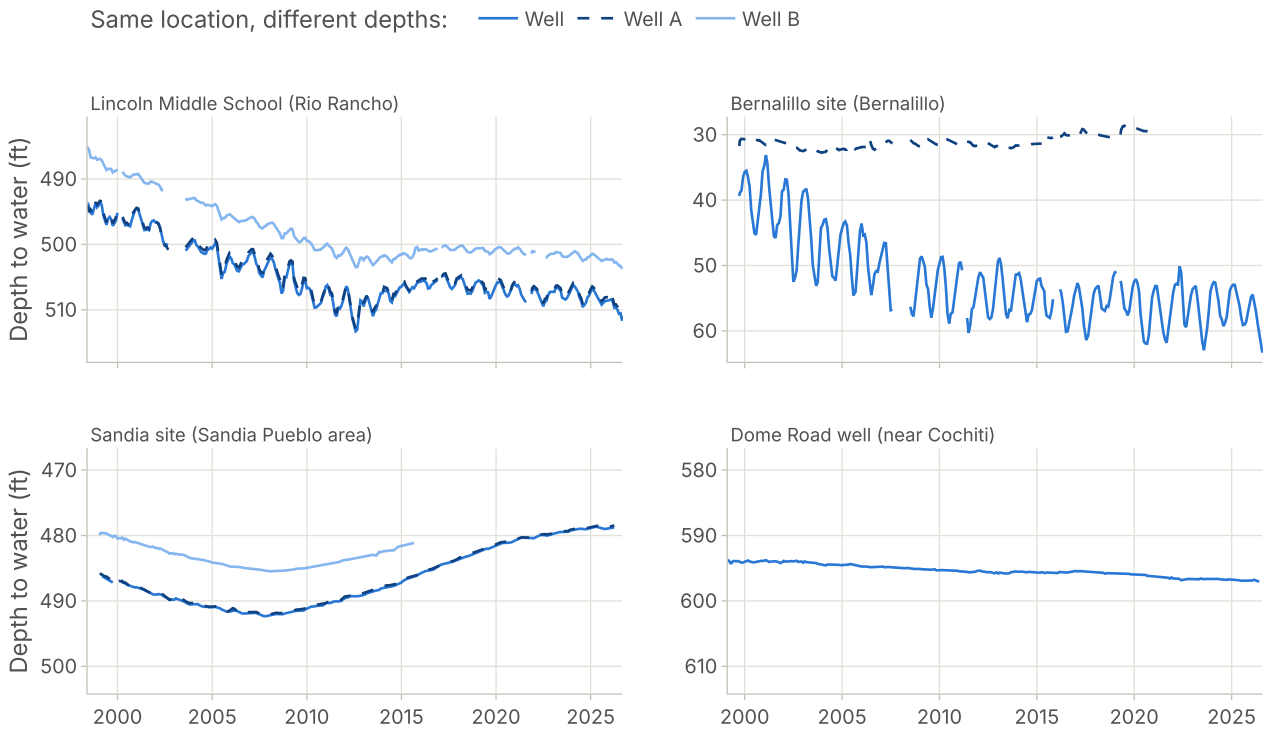


Figure 2. Monthly mean depth to water at the nine USGS monitoring wells, 1998–2026 (axis flipped: up is a higher water table). Wells marked A or B share a location but are measured at a different depth. Every panel spans the same vertical range, so steepness can be compared. Five of the nine wells fell over their records; at the Sandia site the water level has risen about 13 ft since 2007. Scope: Sandoval County, 65–95 km from Cuba (not Cuba). Source: USGS Water Data statistics, parameter 72019, cleaned by J. Wauneka.

which visitors can download. Charts can be zoomed and hovered for exact values. *[Dashboard lead: update this paragraph if the app's pages change before Nov 10.]*

5 Results

5.1 Groundwater Levels

The nine USGS monitoring wells sit at four sites between Rio Rancho and Cochiti, 65–95 km from Cuba (Fig. 2). Over their full records, five wells fell, three rose, and one showed no clear trend. Water levels at Lincoln Middle School (Rio Rancho) fell 0.36–0.40 ft per year and were still falling in 2016–2025, as was the Dome Road well near Cochiti. The Bernalillo site's main well fell about 0.5 ft per year over its record but showed no clear trend in 2016–2025. At the Sandia site the water level fell until 2007 and then rose about 13 ft by 2025 (*Measured*). That recovery coincides with the San Juan–Chama Drinking Water Project, after which the Albuquerque water utility pumped less groundwater and water levels rose across the Albuquerque area (Ritchie et al., 2019); these data alone cannot show the cause (*Hypothesis*).

In the Placitas area, only five NMBGMR wells were measured both in 1997–98 and in 2018–19. Three changed by 5 ft or less (two of them from baselines flagged as recently pumped), one (PC-014) is 51 ft deeper, and one (PC-025) is 57 ft shallower; PC-025's 1990s readings rose steadily after a first reading flagged as recently pumped, so its 1990s average is not a stable baseline. Five wells cannot establish an area-wide trend (*Insufficient data*).

5.2 Wells and Aquifer Units

Where water is deeper, wells are drilled deeper (Spearman $\rho = 0.63$–0.64 within the two larger record types; Fig. 3), but the spread is wide: half of 180 wells have water between 41 and 241 ft below land. Among recorded wells, depth to water was smallest around Cochiti–Peña Blanca and largest around Bernalillo–Rio Rancho, though the groups mix record types and decades. Aquifer properties vary widely: within three formations, transmissivity values span more than the spread between formation medians, up to almost five orders of magnitude in the Madera Limestone. Only the aquifer-test file records the formation, so the other well records cannot be assigned to an aquifer unit (*Insufficient data*).

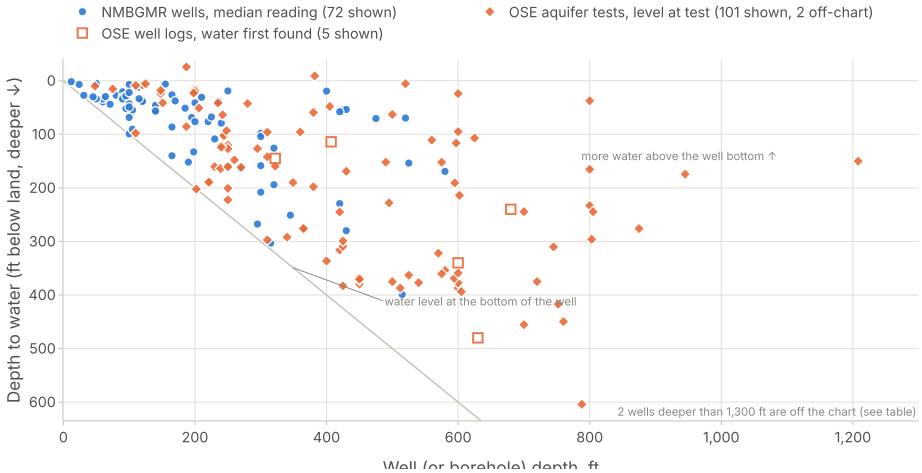


Figure 3. Well depth against depth to water, one point per well, from three record types, 1930–2022. The record types measure water differently (a 1997–2019 reading, the level on the day of an aquifer test, and the depth where water was first found during drilling), so they are context rather than one comparable series. Scope: Sandoval County, including 7 wells within 25 km of Cuba. Sources: NMBGMR groundwater levels; NM OSE aquifer tests and well logs.

5.3 Cuba Case Study

Within 25 km of Cuba, the collected data contain 15 wells with one-time aquifer tests (1957–1979), one well log (2018), and the Rio Puerco stream gage. Nine of the tested wells recorded a depth to water on test day, but no well near Cuba has been measured repeatedly; the nearest well with repeated water-level measurements is 38 km away, and the nearest USGS monitoring well 65 km (Fig. 4). The data therefore cannot confirm or rule out severe decline in Cuba's aquifer (*Insufficient data*). Answering RQ3 requires Cuba's municipal well records: well IDs, producing formations, pumping volumes, and water-level histories.

On the USGS map of the Ojo Alamo Sandstone (Thorn et al., 1990), Cuba sits at the southeastern end of the formation's outcrop, where it is shallow, and its top deepens northward past 1,000 and 3,000 ft. Only eight aquifer-test events in the county name the Ojo Alamo (seven) or the San Jose Formation (one), all from 1953–1977: too few to describe either formation, let alone compare them (Fig. 5). The single San Jose test (100 ft^2/day) is close to the one published Cuba Mesa Member value near Lindrith and within a San Jose range from earlier tests, both compiled by Blandford et al. (2025).

As surface-water context, not evidence about groundwater, the Rio Puerco at Cuba had measurable flow on all four visits in April and May and on none of the six visits from August to March, 2019–2026 (U.S. Geological Survey, 2026b). Spot visits cannot show how often the river runs dry.

5.4 Monitoring Coverage and Data Gaps

Monitoring is concentrated along the Rio Grande: 86% of the 271 mapped sites lie east of 106.75° W (Fig. 1). Groundwater levels have been monitored every year only since 1999, and only at four sites in the east of the county; the collected data have no water-level monitoring in the Cuba area (Fig. 6). According to the dataset briefs, no dataset gives more than limited support for RQ3, the question about Cuba itself.

5.5 Groundwater-Stress Indicators

Five indicators can be computed from the collected data without overstating their certainty: the 2016–2025 water-level trend at the monitored wells (four of the seven wells with enough recent data were falling and two rising), the full-record water-level change at the same wells (−16 to +8 ft), the distance from Cuba to the nearest repeated water-level record (38 km), the share of Rio Puerco visits with flow (4 of 10, river only), and the number and age of aquifer tests near Cuba (15 wells, 1957–1979). Each applies only where its data were collected, and none measures how much groundwater is left; the Streamlit app lists them with their scope and confidence (RQ6).

6 Discussion

[What the results mean for Sandoval County and Cuba, and which indicators best communicate groundwater stress (RQ6).]

7 Limitations

The project proposal included three questions that this report does not answer, because the data needed for them were not obtained during the project:

- **Groundwater quality and salinity.** No groundwater-quality records were collected. The water-quality measurements in the USGS file all come from streams,

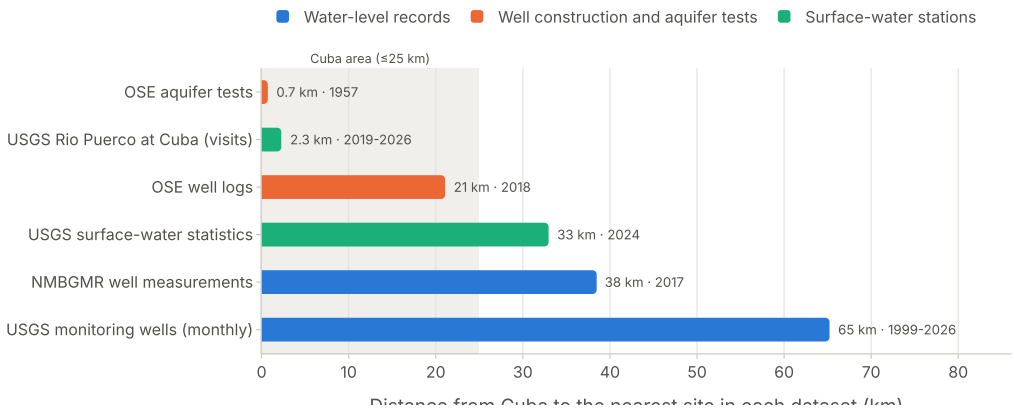


Figure 4. Distance from Cuba to the nearest site in each dataset. The nearest well with repeated water-level measurements is 38 km away; within 25 km there are only one-time aquifer tests (1957–1979), one well log, and the Rio Puerco stream gage. Distances are straight-line from the approximate village center. Sources: USGS monitoring wells and surface-water statistics, NMBGMR groundwater levels, NM OSE aquifer tests and well logs, and USGS station 08332525.

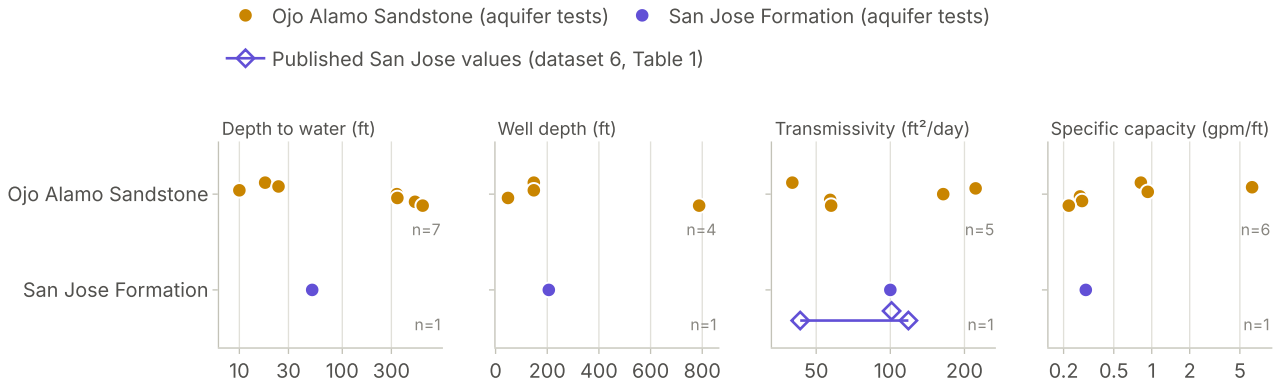


Figure 5. Every aquifer test that names the Ojo Alamo Sandstone (seven events) or the San Jose Formation (one), 1953–1977, with published San Jose transmissivity values compiled by Blandford et al. (2025) (open diamonds). Depth to water, transmissivity, and specific capacity use log scales. Scope: tests 1–45 km from Cuba. Source: NM OSE aquifer tests.

mainly the Rio Grande below Cochiti Dam, far from Cuba.

- **Acequias.** No data on acequia locations or service areas were collected.
- **Groundwater and surface-water links.** Comparing the two requires repeated water-level measurements near a stream. The only stream records at Cuba are 10 visits to the Rio Puerco gage, and the nearest well in the water-level datasets is 38 km away.

The data that were collected also limit what the results can support:

- Almost nothing measures Cuba's groundwater: the nearest well measured more than once is 38 km away, so RQ3 cannot be answered with these data.
- Four monitoring sites carry the county's trend results; they are indicators at those points, not a county average or a measure of remaining supply.
- Most aquifer tests are decades old and record conditions on the day of the test, and only eight name the Ojo Alamo Sandstone or San Jose Formation.
- No geologic-region layer was available, so wells are grouped by nearest town and by the formation named in aquifer tests.
- Trend tests assume independent years; serial correlation makes them look more certain than they are.

8 Conclusions

[Summary of findings and recommended next steps.]

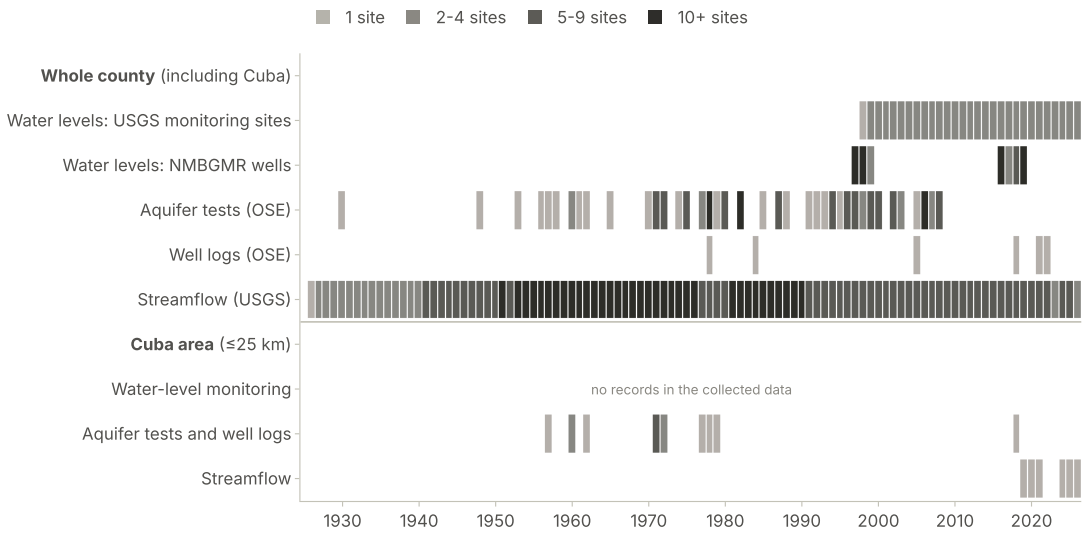


Figure 6. Number of sites with at least one record in each year, 1926–2026, for the whole county (top) and the Cuba area within 25 km (bottom). Darker shading means more sites. Gaps are years with no record in the collected files; other agencies may hold data. Sources: USGS, NMBGMR, and NM OSE.

Data and Code Availability

Code and data
github.com/pdjordj11/eng_220_aquifer_project
Interactive dashboard
eng220aquiferproject.streamlit.app
Documentation
pdjordj11.github.io/eng_220_aquifer_project

Author Contributions

P. Djordjevic: documentation, final report, presentation, and video.
D. Gonzales: data acquisition: collected seven datasets and wrote a research brief for each, plus a plotting guide comparing them.
J. Wauneka: data cleaning: wrote the MATLAB script that produces the cleaned groundwater-level and aquifer-test files.
M. Watkins: analysis code and visualizations: wrote the analysis package (data loaders, trend statistics, and 15 charts with labeled findings) that feeds the app, this report's figures, and the documentation site.
M. Yousef: GitHub workflow and the Streamlit app.

Acknowledgments

[To be written.]

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